

EXPERT OPINION

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Mavrilimumab, a human monoclonal GM-CSF receptor- α antibody for the management of rheumatoid arthritis: a novel approach to therapy

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Introduction: Mavrilimumab, formerly known as CAM-3001, a GM-CSF receptor- α antibody, is the first human monoclonal antibody to be used in Phase II studies (2011) to modulate the innate immunity pathway targeting GM-CSF signaling in moderate rheumatoid arthritis (RA).

Areas covered: Analysis of available clinical trial data on GM-CSF receptor- α antibody and medical literature search using MEDLINE for molecular mechanisms of pathogenesis of RA and its treatment forms the basis of this expert opinion review. The mavrilimumab Phase II double blind, randomized, placebo-controlled ascending dose trial demonstrated statistically significant achievement of primary and secondary end points in patients with moderate RA. The trial demonstrated significant clinical benefit in the 100 mg mavrilimumab cohort compared to the placebo group.

Expert opinion: The novel molecular targeting mechanism of mavrilimumab together with its demonstrated clinical efficacy, tolerability and safety profile in Phase II clinical trials in moderate RA, suggests significant potential utility for this drug to induce clinical remission, reduce flares and improve morbidity and mortality in patients with RA.

Keywords: GM-CSF, mavrilimumab, rheumatoid arthritis

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1. Introduction

Rheumatoid arthritis (RA) is an autoimmune chronic inflammatory disorder. Although the hallmark is a symmetrical polyarthritis of synovial joints, this is a systemic disease with many extra-articular manifestations including disease in eyes, heart, lungs and skin. The condition is mediated by dysregulation of immune cells, which invade the synovium and lead to the formation of an inflammatory pannus and subsequent joint destruction. The combination of inflammation and cardiovascular risks are associated with a significant decrease in life expectancy [1,2]. Overall, RA affects approximately 0.5 – 1% of the population worldwide [3], although there is considerable regional variation [4,5]. It is two to three times more common in women and while potentially affecting all ages, the incidence peaks between the ages of 55 – 64 years [6,7].

Data from the Rochester Epidemiology Project showed that, between 1995 and 2007, 41 per 100,000 people were diagnosed with RA each year. Although there was a small decrease in incidence (0.5%) among men, this was offset by an overall increase in rate of 2.5% each year [6], indicating a growing burden of disease on society.

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Box 1. Drug summary.

Drug name	Mavrilimumab
Phase	Phase II clinical trials
Indication	Rheumatoid arthritis
Pharmacology description	Anti-GM-CSFR Monoclonal antibody
Route of administration	Injectable, subcutaneous
Chemical structure	Biological, protein, recombinant IgG4 antibody
Pivotal trial	NCT01050998

As RA is associated with considerable morbidity, disability, decreased quality of life and mortality, it has large socioeconomic implications on society [8-13].

1.1 Pathogenesis

Although the pathogenesis of RA has yet to be fully elucidated, recent years have witnessed an explosion in the understanding of basic science pathophysiological processes that appear to underlie it. Genetic [14-17] and environmental factors [18] combine to create a milieu of RA susceptibility that, when as yet unknown triggers are encountered, provokes the immune system to activation in a dysregulated manner with heterogeneous molecular and cellular interactions. The complexity of this process provides both challenges to management of the disease and new opportunities. For example, although the proinflammatory cytokine TNF- α was thought to be central to the disease process in RA, neutralization of this molecule with biologic agents in the clinic is only effective in some 60% of cases.

A more sophisticated analysis of immune function in RA suggests a complex interaction between cells of innate and adaptive immune systems. Activation of the innate immune system through various stimuli leads to activation of adaptive immune system via presentation of antigen to T cells, which in turn can stimulate other cells such as monocytes, macrophages, polymorphonuclear cells and synovial fibroblasts to release further inflammatory and toxic mediators including prostaglandins, nitric oxide, free radicals, proinflammatory cytokines and chemokines, granulocyte-macrophage colony-stimulating factor (GM-CSF), interleukin-6 (IL-6), IL-1, IL-8, matrix metalloproteases (MMP) and monocyte chemo-attractant protein (MCP-1), which will in turn attract other inflammatory cells to the site of inflammation or mediate tissue damage. One previously overlooked, but now emerging as a potentially important aspect of the inflammatory process underlying RA is the role of the neutrophil [19]. These cells not only have the ability to modulate the immune system, they also confer a huge potential to damage tissue by secreting an especially potent array of reactive oxygen intermediates and proteases.

1.2 GM-CSF and role in RA

The hematopoietins are a family of polypeptide growth factors involved in the regulation and proliferation of bone

marrow stem cells, directing the differentiation into monocyte-macrophages, erythrocytes, granulocytes, megakaryocytes and lymphocytes. The lineage-specific development and differentiation of the monocyte-granulocyte system is dependent on growth factors called the granulocyte-macrophage colony-stimulating factors (GM-CSF) [20].

GM-CSF is a pleuripotent cytokine, which regulates the functions of myeloid cells involved in innate immunity, including neutrophils and macrophages. Following priming by GM-CSF and activation by appropriate immune complexes, neutrophils can secrete pro-inflammatory cytokines including IL-1, IL-8, TGF β , TNF- α , express class II MHC and present antigen to T cells, further contributing to the inflammatory process. Activated macrophages release pro-inflammatory cytokines TNF- α , IL-6, IL-1 and various chemokines.

Evidence for a key role of GM-CSF in RA [21,22] is found in many studies [23,24]. Singh *et al.* demonstrated early macrophage infiltration in RA synovium in early RA, even prior to the significant development of lymphoid follicles or vascularity [23]. Xu *et al.* described the abundant production of GM-CSF in rheumatoid synovial effusions [24] and another study had demonstrated high titers of GM-CSF in moderate-to-severe RA compared to controls [25]. In active inflammation, there is increased expression of GM-CSF in plasma and the synovium, which plays an important role in differentiation and function of macrophages and neutrophils. Bresnihan *et al.* found a correlation between synovial lining macrophages and articular destruction in RA [26]. The changes in the number of CD68⁺ synovial sublining macrophages with anti-rheumatic therapy in clinical studies correlated with improvement in DAS28 score and therefore, could potentially serve as a biomarker for therapeutic response to biologic and non-biologic DMARDs [27-29]. Animal studies have showed accelerated inflammation [30] and exacerbation of collagen-induced arthritis with GM-CSF [31].

GM-CSF also plays a role in osteoclastogenesis both directly and also through regulation of macrophage colony-stimulating factor (M-CSF). Osteoclast precursors express receptor activator of nuclear factor-K β (RANK) and develop into functional osteoclasts in the presence of M-CSF and RANKL [32]. This, along with the cytokines in the inflammation-rich RA joint, contributes to osteoclastic bone resorption, ultimately leading to joint damage. Production of GM-CSF, induced by cytokines IL-1 and TNF- α , is thought to have a role in proliferation of pannus and joint destruction [33].

2. Overview of the market—current scenario in RA therapy**2.1 The unmet need**

There is considerable morbidity and mortality still associated with RA. Despite the availability of effective DMARDs and biologic therapy, many patients do not tolerate or respond to treatment, leading to persistent inflammation, joint

destruction and cardiovascular morbidity. The development of additional therapeutic strategies are therefore required, ideally supported by biomarkers to identify potential responders, thereby preventing progressive joint damage and improving quality of life.

2.2 Currently licensed biologic therapies for RA

Tumour necrosis factor inhibitor (TNFi) drugs specifically target TNF- α , a pleiotropic proinflammatory cytokine secreted by various cells including macrophages, neutrophils, monocytes, T-cells, B-cells, NK cells following stimulation by interleukins, GM-CSF, interferons and various cytokines. Current biologic TNFi drugs include infliximab (a chimeric IgG1 κ monoclonal antibody specific for human TNF- α); adalimumab (recombinant human IgG1 monoclonal antibody specific for human TNF- α); etanercept (a fusion protein comprising two human p75 TNF receptors linked to the Fc portion of human IgG1 and also binding lymphotoxin); certolizumab Pegol (a recombinant pegylated humanized anti-TNF antibody) and golimumab (human IgG1 κ monoclonal antibody specific for TNF- α). These agents are usually used as first-line biologics and have a high efficacy in up to two-thirds of patients with RA [34].

2.2.1 Rituximab

Rituximab is a chimeric human immunoglobulin G1 (IgG1) anti CD-20 antibody that selectively depletes CD20 expressing B cells. The therapeutic success of this agent in RA highlights a key (but as yet not fully defined) role that B cells play in RA pathogenesis. One cycle of two infusions of rituximab may produce clinical benefit of decreased disease activity of many months [35]. Repeated cycles appear to be tolerated well. Rituximab is currently used mainly in situations of TNF failure or intolerance, but also has merit as first-line biologic.

2.2.2 Tocilizumab

Tocilizumab is a humanized monoclonal IgG1 antibody that targets and blocks the interleukin-6 receptor and thereby inhibits the effects of IL-6, one of the most abundant pro-inflammatory cytokines in the rheumatoid synovium. IL-6 is a pleiotropic cytokine, which activates neutrophils and macrophages. Tocilizumab improves treatment outcomes in RA patients, may be used as first- or second-line biologic and, in contrast to other biologic therapies, is effective in monotherapy without the requirement for co-prescription of methotrexate [36,37].

2.2.3 Abatacept

Abatacept is a soluble fusion protein that consists of a modified Fc portion of human immunoglobulin G1 (IgG1) fused to the extracellular domain of human cytotoxic T-lymphocyte-associated antigen 4 (CTLA-4). It is a selective co-stimulation modulator that inhibits T-lymphocyte activation by binding to CD80 and CD86, thereby blocking interaction with CD28, a signal required for full activation of T lymphocytes.

Trials have shown improvement in RA symptoms in patients who were inadequate responders to anti-TNF therapy [38].

2.3 Introduction to anti-GM-CSF therapy

GM-CSF receptors are heterodimers consisting of a ligand-specific α -subunit (GM-CSFR- α) and a β -chain subunit that is shared with the IL-3 and IL-5 receptors. The α -receptor binds to the ligand with low affinity, and confers stability to the complex with β -chain subunit; the common β subunit is associated with Janus Kinase 2 (JAK2) activation, tyrosine kinase activity and signal transducer and activator of transcription (STAT1/STAT3/STAT5) intracellular signaling cascades.

GM-CSF and its receptors are abundantly found in the synovial fluid, synovial tissue and plasma of patients with RA. GM-CSF-differentiated CD68⁺ macrophages in RA synovial joints are potential biomarkers for treatment response [27-29] and observations that administration of GM-CSF in Felty's syndrome to correct granulocytopenia resulted in arthritis flares [39], suggested the potential role of this cytokine in RA. Therefore, it was hypothesized that inhibiting the critical cytokine signaling pathways of GM-CSF through GM-CSFR blockade will have significant anti-inflammatory effects.

Mavrilimumab (CAM-3001) (see Box 1) is a novel high-affinity human monoclonal IgG4 antibody against GM-CSFR- α and a competitive antagonist of GM-CSF signaling (Figure 1). It has been the first studied GM-CSFR antagonist in RA and first agent targeting this innate immunity pathway of RA pathogenesis.

3. Drug development

3.1 Preclinical studies

In vitro studies demonstrated that GM-CSF stimulated peripheral blood mononuclear cells (PBMC) to produce TNF in a dose-dependent manner. The GM-CSFR antibody, CAM-3001 inhibited IL-8, IL-6 and TNF in a dose-dependent manner. In mouse models of collagen-induced arthritis, anti-GM-CSF antibodies decreased the severity and duration of arthritis (data presented at the 70th Annual ACR Meeting in Washington DC, November 2006).

This favored the hypothesis that GM-CSF inhibition would block important and critical cytokine pathways of RA pathogenesis [40]. *In vivo* studies in Cynomolgus monkeys, CAM-3001 inhibited or attenuated GM-CSF induced leucocyte margination and leucocytosis [41]. Mavrilimumab demonstrated equipotency in GM-CSFR blockade in humans and cynomolgus monkeys.

3.2 Phase I trial

In the Phase I study (NCT00771420) (Table 1), 32 subjects aged 18 – 70 years who had mild-to-moderate RA (DAS 28 score $\leq$ 4.8) for more than 6 months, satisfying ACR criteria for RA and who had methotrexate treatment for > 3 months were given single, escalating intravenous

Mavrilimumab in RA

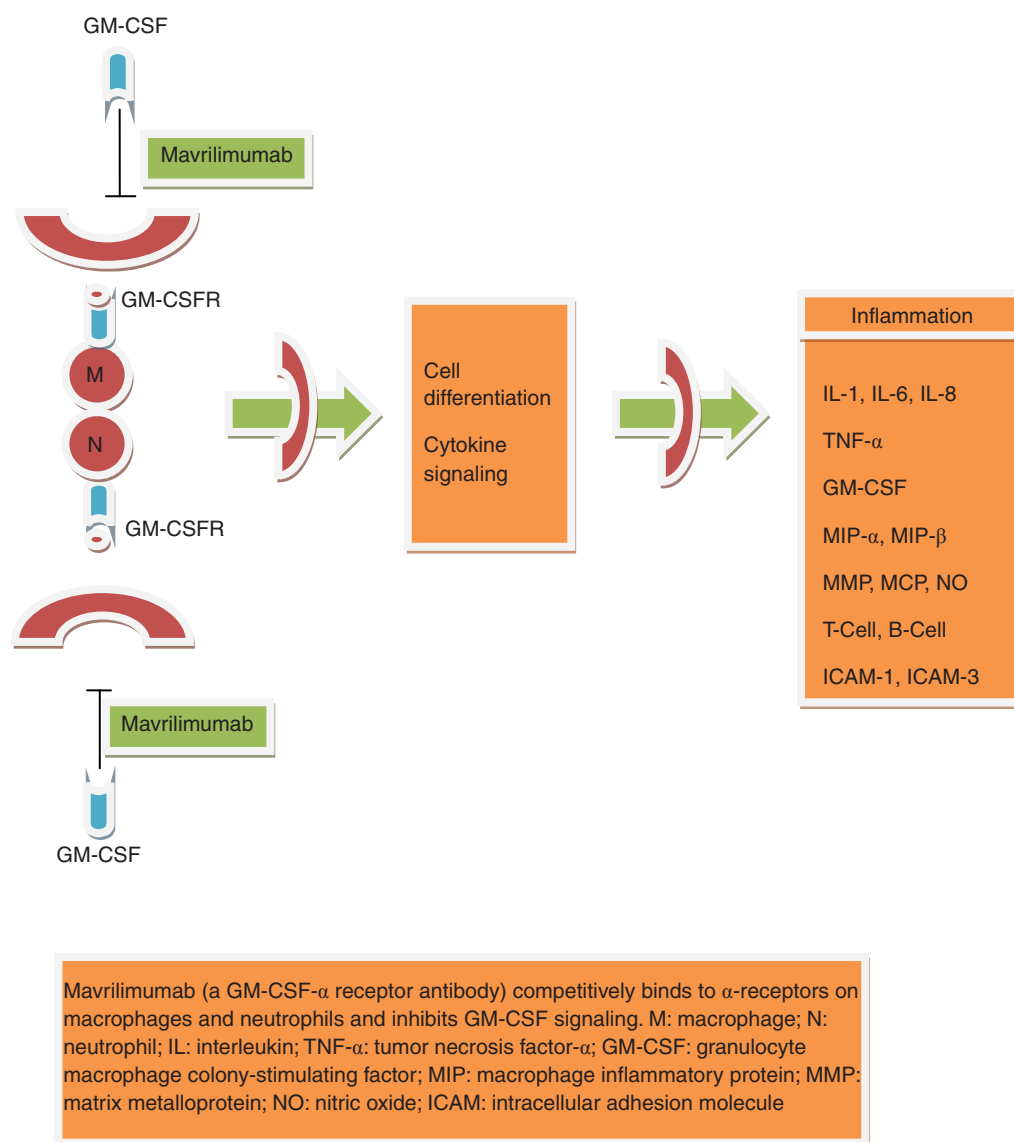


Figure 1. Mavrilimumab (a GM-CSF- α receptor antibody) competitively binds to α -receptors on macrophages and neutrophils and inhibits GM-CSF signaling.

GM-CSF: Granulocyte macrophage colony-stimulating factor; ICAM: Intracellular adhesion molecule; IL: Interleukin; M: Macrophage; MIP: Macrophage inflammatory protein; MMP: Matrix metalloprotein; N: Neutrophil; NO: Nitric oxide; TNF- α : Tumor necrosis factor- α .

dose of mavrilimumab (0.01, 0.03, 0.1, 0.3, 1.0, 3.0 and 10.0 mg/kg) or placebo on study day 0 to assess the safety, tolerability, pharmacokinetics and pharmacodynamic profile [42].

At doses of 0.1 and 10 mg/kg, the peak plasma concentration (C_{max}) was 0.241 and 373 (μ g/ml), respectively. Systemic clearance (CL) decreased from 88.8 to 3.15 ml/kg/day when the dose was increased from 0.01 to 10.0 mg/kg. The elimination half-lives at 1.0 and 10 mg/kg were 4.78 and 15.0 days, respectively. The distribution volume was not ascertained due to non-linear pharmacokinetics at lower doses. The clearance of mavrilimumab was thought to be mediated by

two pathways; first, elimination by the reticulo-endothelial system and second by the target antigen receptor-mediated clearance or the antigen-sink effect [43].

Statistically significant inhibition of suppressor of cytokine signaling 3 (SOCS3) mRNA by mavrilimumab was noted at doses of 1.0 mg/kg, with effects sustained for up to 4 weeks, suggesting blocked expansion of peripheral white blood cells.

Mavrilimumab generally appeared to be well tolerated. The side effect (SE) profile was mild and unrelated to drug dosage; the most commonly reported SEs were headache, nausea, vomiting, diarrhea, peripheral oedema, nasopharyngitis,

Table 1. Clinical trials (summary) of mavrilimumab in RA.

Trial (status)	Drug regime CAM-3001 (Mavrilimumab)	Primary outcome measures	Secondary outcome measures	Clinical profile	
				Adverse effects (AE)	CRP/ESR DAS 28
Phase I (completed) NCT00771420 Randomized, double-blind, placebo-controlled	Escalating single doses of mavrilimumab (mg/kg) 0.01 mg/kg, 0.03, 0.1, 0.3, 1.0, 3.0, 10.0 and a placebo	Incidence and severity of adverse events	Pharmacokinetic parameters of CAM-3001	Minor AE profile—no relation of dose to severity of AE; One episode of urticaria of face and neck; No serious AE	Reduced CRP, ESR and DAS-28 at 4 weeks
Phase II (recruiting) NCT01050998 Randomized, double-blind, placebo-controlled, ascending dose	Mavrilimumab 10, 30, 50, 100 mg subcutaneous injections every other week and a placebo	Improvement in DAS-28 CRP	ACR20, ACR50, ACR70, DAS28 responses	Minor AE: change in DLCO, nasopharyngitis, pharyngitis, minor rise in transaminases, upper respiratory infection Serious AE: unrelated to treatment	Improved CRP and DAS 28 responses; Improved ACR20, 50, 70 responses

paraesthesia and back ache. No significant hematological toxicity was noted. One episode of significant face and neck urticaria was noted in a subject at 10 mg/kg. No abnormalities of lung function were noted despite worries about potential effect on surfactant clearance.

Exploratory analysis in subjects with DAS 28 score > 2.6 showed remission at week 4 and those with moderate activity showed statistically significant DAS 28 responses. The CRP and ESR also showed improvement at 4 weeks.

3.3 Phase II trial

Burmester *et al.* conducted a Phase II (NCT01050998) (Table 1), randomized, double-blind, placebo-controlled, multiple drug dosing trial in patients with RA with moderate disease activity (DAS 28 score > 3.2) who were on stable doses of methotrexate. In this dose-ranging study, 216 subjects (who completed the study) were randomized to receive 10, 30, 50 or 100 mg of subcutaneous mavrilimumab or a placebo. The results were presented to the American College of Rheumatology (ACR) conference in 2011 [44].

For the DAS 28-CRP responses of > 1.2 at day 85, the primary end points, were 55.7% compared to 34.7% in the placebo ($p = 0.003$) group, reflecting a 21% difference, with a mean change in DAS 28 score of -1.51. The response rate was 41 and 66.7% in the 10 and 100 mg group, respectively, suggesting higher dose-dependent responses. The DAS 28-CRP remission was 23.1% in the 100 mg group compared to 6.7% in the placebo group ($p = 0.016$), with remission responses again improving in a dose-dependent manner.

The ACR 20, 50 and 75 responses were significantly better in the mavrilimumab group compared to the placebo, being statistically significant in the 100 mg group ($p = 0.005/0.021/0.030$, respectively). The HAQ-DI (health assessment questionnaire-disability index) improvement, measured as an improvement of ≥ 0.25 , was again

significantly better in the mavrilimumab group ($p = 0.009$). The onset of clinical response generally occurred within 2 weeks.

Of the notable adverse effects (AE), a decrease in diffusion capacity for CO (11.9%), nasopharyngitis (6.3%), pharyngitis (2.5%) and increased transaminases (2.5%) were higher than the placebo group; however, the adverse event profile was mild and there were no significant toxicity alarms. Four serious adverse events (SAE) unrelated to the treatment were reported.

The Phase I/II trials demonstrated efficacy with a mild AE profile (trial summary table 1).

3.4 Ongoing studies

Ongoing clinical study includes NCT01023256, a multi-center, randomized, double-blind, placebo-controlled Phase Ib/IIa trial of MOR 103 (MorphoSys AG), a fully human monoclonal antibody (mAb) directed toward human GM-CSF, in patients with RA with moderate disease activity (DAS 28 ≤ 5.1).

4. Expert opinion

The development of the GM-CSF receptor- α monoclonal antibody has the potential to be a significant new development in the armamentarium of biological drugs available to treat RA. Despite the successful introduction of targeted biologic therapies for therapy in RA, a considerable unmet therapeutic need remains, as many patients do not respond to other biologic therapies due to primary or secondary resistance or experience intolerable SEs.

The unique mechanism of action of mavrilimumab as a GM-CSFR antibody results in inhibition of what may be a central pathogenetic pathway underlying inflammation in RA, namely the inhibition of macrophage and neutrophil activation.

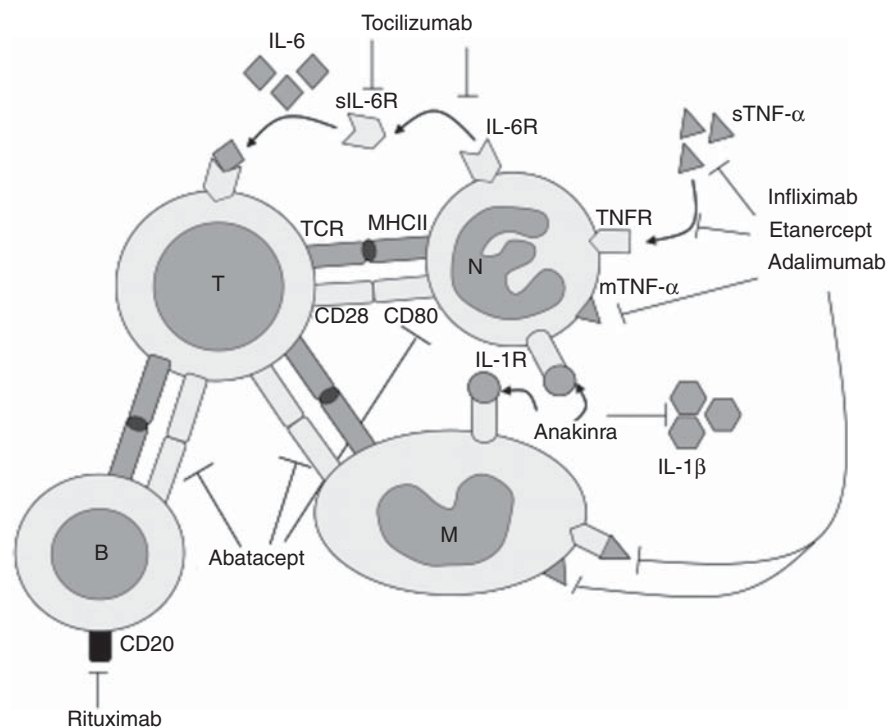


Figure 2. Summary of the immune targets of biologic therapies. Anti-TNF drugs infliximab, etanercept and adalimumab target soluble (sTNF- α) and membrane-expressed TNF- α (mTNF- α), facilitating blockade of signaling and removal of mTNF- α -expressing cells. Anakinra is an IL-1 receptor (IL-1R) antagonist, which competitively binds the IL-1R and inhibits IL-1 signaling. Abatacept prevents the co-stimulation of T cells by blocking engagement of CD28 with CD80/86, preventing signaling from MHCII on antigen-presenting cells to the TCR. Rituximab targets and removes CD20⁺ B cells. Tocilizumab inhibits IL-6 signaling by blockade of the soluble (sIL-6R) and transmembrane (IL-6R) IL-6 receptors.

Reproduced with permission from [19].

B: B cell; M: Macrophage; N: Neutrophil; T: T cell.

Although the clinical studies available are as yet only few, the results to date, while preliminary, are reassuring. The Phase I trial has reported good tolerability and safety of the drug with pharmacodynamically important inhibition of SOCS3.

The Phase II trials showed good DAS28 CRP, ACR20, ACR50 and ACR70 responses, with beneficial clinical responses occurring within 2 weeks and maintained at 12 weeks compared to placebo. There were also notable effects on function, with improvements in HAQ-DI. The SEs appeared minor and no reports of additional SAE, including anaphylaxis, related to treatment were encountered. An anticipated signal of potential lung toxicity due to inhibition of macrophage clearance of lung surfactant was not observed.

While available data, as expected in early-phase drug development are limited, they are encouraging. Further work, with pivotal studies will of course be required, focusing on developing further efficacy and safety data, as well as determining effects on structural progression and even, potentially, head-to-head studies with established biologics. In our opinion, mavrilimumab has clear potential utility as therapy for RA. It is too early to anticipate where it might fit within

the treatment algorithm, whether post-biologic failure or as first-line biologic therapy. Given the emerging potential key role for innate cellular immunity in RA, the ability of mavrilimumab to suppress inflammation by inhibiting this route offers much promise. It is likely to be effective in reducing radiographic progression, potentially comparable to anti-TNF drugs and IL-6 inhibitors, through inhibition of proliferation of inflammatory pannus and osteoclastogenesis. Theoretically, the potential for mavrilimumab in RA is high, comparable to IL-6 and TNFi (Figure 2); however, long-term efficacy data will be needed to substantiate it.

At present the dose of 100 mg of subcutaneously every 2 weeks appears to be the most effective dose with good safety profile during the limited duration of current studies. However, the long-term safety data, in a larger group of patients, including potential for reactivation of latent mycobacterial infection, effect on infection, in general, and potential for malignancy needs to be available before this can be definitely advocated given the diverse immunomodulatory effects of GM-CSF. The Phase II trial was mainly European: further evaluation in a wider population is required to assess efficacy and AE profile. In addition, use as monotherapy should

be evaluated, particularly in patients who are intolerant of methotrexate.

Further studies are also needed to assess the full potential for its use outside suppressing synovial inflammation in RA. For example, inhibition of SOCS3 may offer opportunities to target metabolic aspects of chronic inflammation. The SOCS are a family of proteins expressed in response to cytokines, which interact with cytokine receptors to switch off their activity. Their constitutive expression in chronic inflammation is implicated in insulin resistance, obesity and metabolic syndrome.

Trials to date report a favorable outcome in RA patients with moderate disease activity as early as 2 weeks and sustained at 12 weeks. However, the real test may be the long-term response in high disease-activity RA cohorts. The population of patients who may benefit from mavrilimumab will become clearer in future when further robust studies are conducted, more data becomes available and novel biomarkers to target therapy are

identified. The next 5 years of clinical development will be instrumental in addressing these important questions and determine if this interesting new agent is able to live up to its promise.

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Declaration of interest

The authors declare no conflicts of interest and have received no payment in preparation of this manuscript.

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